Verify, Then Vote: Overturning Wrong Majorities in Multi-Agent LLMs

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November 2026

Abstract

Checking agents' cited passages improves majority vote, but an accurate aggregate can still overturn correct answers. We study *verify, then vote*: a fixed plurality default may be overturned only by an answer whose passage an accepted checker labels as supporting it. On OVERTURN's 1,000 primary questions, the pre-specified scored rule raises accuracy from 77.1% to 85.0%; on 526 decisive conflicts, from 71.5% to 86.5%. It nevertheless misses its pre-specified harm criterion: the upper confidence bound on overturns of correct majorities is 5.5%, above the 5% ceiling. *Verified default first*, which keeps a verified default, was developed after this analysis and specified before a second agent pool answered the same questions, partly reusing checker labels. It reaches 85.9% on that pool's decisive conflicts, with a harm-rate upper bound of 2.2%. This tests transfer across agent pools, not generalization to unseen questions. Matched-parameter ablations isolate the gate's reduction in harmful overturns, and withholding or replacing cited passages removes most corrections. The formal property identifies an inspectable failure condition: losing a correct default requires a verified wrong answer; it does not establish that such answers are rare. A strong model judge is more accurate and initially cheaper (93.9% versus 86.5%; \$0.014 versus \$0.021 per contested primary decision). The benefit of verify-then-vote is an explicit selection policy over reusable evidence: all 1,100 benchmark decisions replay byte-identically, and leave-one-agent-out decisions reuse sealed labels without a model call.

Keywords: multi-agent LLM systems, arbitration, evidence verification, majority vote, replay, provenance, preregistration

1 Introduction

When several LLM agents disagree, majority vote is a natural way to combine their answers (Wang et al., 2023). It fails when the agents share a mistake. The passages they cite may supply corrective evidence, but citations alone do not establish support, and overturning a majority can introduce a new error. We study how to use that evidence while keeping each overturn auditable.

We study a common process for combining these answers. Each agent submits an answer and cites passages it used. A checker labels each (answer, passage) pair once as supports, refutes, or not-enough-information. We record the answers, passages, and labels, then apply a fixed selection rule to those records. The rule starts from a plurality default and permits an alternative to replace it only when an accepted checker label verifies that alternative. We call this process *verify, then vote*.

The rule variants differ in how they select among verified answers. *Verified vote* takes the most common verified answer; the *scored rule* instead uses recorded evidence and confidence. *Verified default first* keeps the default whenever it is verified and otherwise uses verified vote. We distinguish these rules from the reported majority baseline, whose tie-breaking can differ from their defaults. The formal harm property compares each rule with its own default; the empirical evaluation also compares it with the reported majority baseline.

A model judge is the stronger choice in our experiments if first-decision accuracy and cost are the only objectives. The case for verify-then-vote is an inspectable selection policy over reusable evidence. For example, an auditor investigating one agent can remove its answers and recompute decisions from the recorded passages and labels, without another model call. A judge can provide citations and explanations too, but obtaining its judgment on changed inputs generally requires a new call. Reuse here is restricted to existing sealed inputs and accepted checker labels; new evidence may need new checking.

The rule can lose a correct default only by selecting a verified wrong answer. This identifies a place to inspect the failure, rather than certifying checker or source reliability. Practical harm must be measured separately.

Contributions.

1. **Auditable selection.** We specify the verification gate, prove its necessary condition for harm, and record the passage labels licensing each overturn (Section 3).
2. **Controlled evaluation.** We distinguish overall and conflict accuracy, isolate the gate through matched ablations, and test passage dependence and transfer across agent pools. The original scored rule misses its harm criterion; the second pool reuses questions and some labels (Sections 4.1–4.4).
3. **Reusable evidence.** We measure replay, tamper detection, and the cost of re-decisions over sealed inputs, including leave-one-agent-out counterfactuals (Sections 4.5 and 4.6).

The accompanying artifact contains the analysis scripts, sealed inputs, Lean proofs, and replay tests. Supplement, Section S7 maps results to reproduction commands.

2 Setting and Threat Model

Several agents, typically different LLMs, answer the same question, each with an answer, a stated confidence, and citations to the passages they used. An *arbitration step* decides the combined answer. We want that decision to have three properties an auditor can check. *Replay*: from the recorded inputs, anyone can recompute the same decision bit for bit. *Provenance*: the decision is one of the submitted answers, and every input it read traces to an agent, a captured passage, or a recorded checker label. *An explicit rule*: why one answer beat another is a stated function of recorded inputs, so a changed rule can be re-applied to old decisions and compared.

Before selection, the answers, cited passage captures, checker labels, and policy inputs are recorded in a sealed artifact set. The reducer then reads only those recorded inputs, rather than fetching passages or asking a model again. Content hashes let a verifier detect edits against an independently held commitment, subject to hash-collision assumptions; they do not establish that a passage or label is true. Supplement, Section S9 specifies capture, sealing, and the replay guarantee.

These properties suit applications that need to inspect and revise decisions after their first use. An auditor can inspect the passage that licensed an overturn, remove an agent, or compare selectors using the same sealed inputs. New answers, passages, proposition renderings, or checker identities can require new labels. The experiments therefore compare both first-decision quality and the cost of later permitted re-decisions (Sections 4.1 and 4.5); reproducibility alone does not imply better accuracy.

The primary threat is the *honest-but-fallible* agent, which hallucinates, holds stale knowledge, or is miscalibrated. The case verify-then-vote targets is *correlated error*, where most agents assert the same wrong answer. The trusted computing base is the reducer, the capture tooling, and the checker. A checker can err, and a retrieved source can itself assert a wrong answer. The property in Section 3 identifies these verified wrong answers as necessary for losing a correct

default; it supplies no numerical guarantee on their frequency. An agent may also cite a genuine source that says something other than what it claims. That leaves every hash intact, and it is exactly what the checker is intended to catch.

3 Verify, Then Vote

Majority vote fails when most agents share a mistake, and the agents' citations are the natural signal that can say so. Weighting citations is not enough: many on-topic but silent passages can outvote a correct majority (Section 4.3). Verify-then-vote makes verification, not weight, the only condition under which its fixed plurality default can be overturned.

3.1 The rule

Each agent answers a question with a value and cites passages it was shown. A checker labels each (answer, passage) pair once as supports, refutes, or not-enough-information. The answer is rendered as a one-sentence proposition, and the checker sees only that proposition and the passage. A label counts only if the checker's identity is on the policy's list of accepted checkers. That list is part of the hashed policy. The reducer enforces acceptance-before-use; validation before acceptance is the study or application's responsibility.

For example, suppose three agents answer "Sydney" and two answer "Canberra" to a question about Australia's capital. The plurality default is Sydney. If an accepted checker labels a cited passage as supporting Canberra and no cited passage verifies Sydney, the rule selects Canberra. If neither answer is verified, it retains Sydney. If both are verified, the selector determines which to choose. This example separates the verification requirement from the choice among answers that meet it.

For one question, a *value* is a normalized answer with its polarity, and $n(v)$ is the number of distinct agents asserting v. The plurality set $\mathcal{P}$ holds the values of maximal n. An agent's answer is a *witness* if one of its citations carries an accepted supports label for that answer's own proposition, and the verified set $\mathcal{W}$ holds the values with a witness. A *selector* sel maps a non-empty set of values to one of its members.

Definition 1 (Verify, then vote)**.** The default winner is $D = \mathrm{sel}(\mathcal{P})$. It is deterministic for a fixed selector and policy; a scored selector can use evidence labels to break a plurality tie, so changing its policy can change D. If $\mathcal{W} = \emptyset$, the winner is D. Otherwise it is $\mathrm{sel}(\mathcal{W})$.

A verified minority therefore overturns an unverified majority, and when the majority is itself verified the selector decides among the verified values. The simplest selector is plurality with a fixed tie order, which gives *verified vote*: the most common verified answer, or the most common answer when none is verified. A richer selector scores candidates by their evidence, confidence, and other recorded features and breaks near-ties deterministically. Supplement, Section S2 specifies the scoring reducer we use and proves when its tie-break cannot change correctness. For *verified default first*, first fix a separately named base selector b, compute $D_b = b(\mathcal{P})$, and fix a fallback selector f that also returns a member of its input. Define

$$\mathrm{sel}^{\mathrm{VDF}}_{b,f}(S) = \begin{cases} D_b, & D_b \in S, \\ f(S), & D_b \notin S. \end{cases}$$

Use this selector in Definition 1. Since $D_b \in \mathcal{P}$, its default is D_b, computed from b before the gate is applied. It keeps this default whenever it is verified and otherwise selects among verified alternatives. In the evaluated plurality variant, the default is the reported majority baseline, with its seeded tie rule; the fallback counts verified individual responses and uses a separately fixed seeded tie rule. These are fixed selection conventions, not additional evidence checks. The scored variant uses the scoring reducer for both b and f and retains that reducer's default D.

All three are order-free: the decision depends on the set of answers and labels, not on the order in which they arrived. Each decision records its stage, default winner, candidates, and witnesses, so an auditor sees which label licensed each overturn.

3.2 What the rule can lose

The rule's harm property needs one fact about the selector: it returns a member of its input.

Proposition 1 (Harm requires a verified wrong answer)**.** *Let correctness be a property of the value, so that all answers with the same value are graded alike. If D is correct and the winner is not, then some incorrect value carries an accepted supports label.*

Proof. Let w be the winner. Since w is incorrect and D is correct, $w \neq D$, so $\mathcal{W} \neq \emptyset$ and $w = \mathrm{sel}(\mathcal{W}) \in \mathcal{W}$. Hence w is verified and incorrect.

Corollary 1 (Counting form and no-loss)**.** *Over any set of questions, let c_D count those where D is correct and the winner is not, and F those where some incorrect value is verified. Then every c_D question is an F question, so $c_D \leq F$. On a question where no incorrect value is verified, the winner is correct whenever D is.*

Proposition 1 and Corollary 1 are machine-checked in Lean 4 for any selector with this closure property (Supplement, Section S4). This is an inspectable necessary condition for harm, not a guarantee that verified wrong answers are rare or that the checker is reliable. Their frequency must be measured separately. The results hold for verified vote, for the scoring reducer, and for verified default first alike: the last returns the default or a member of its input, so selection closure holds with no new proof obligation.

A verified wrong answer arises in three ways. The checker can label a passage as supporting an answer it does not support, a *checker false support*. The passage can be accurate about a different entity with the same name, so that it supports the answer only under the wrong reading of which entity is meant, an *entity mismatch*. Or the passage can itself state a false or outdated fact about the intended entity, a *false source*. A passage stating that Sydney is the capital of Australia supports "Sydney," a correct checker rightly labels it so, and the gate can then overturn a correct "Canberra" majority. The theorem covers all three; checker validation measures the first, entity resolution and proposition rendering govern the second, and the corpus governs the third. In evaluation, a fourth source counts as a verified wrong answer: gold that misses an acceptable answer. Section 4.2 classifies all four on OVERTURN.

4 Evaluation

We report accuracy, harmful overturns, passage controls, transfer to a second agent pool, and the cost and audit value of replay. Negative results and the limits of each comparison accompany the corresponding findings.

Evaluation scope and methods. OVERTURN contains 1,000 primary questions (set P: SciFact, PopQA, and FEVER) and a separate 100-question TruthfulQA stratum (T). Five small open-weight agents answer using retrieved passages from a fixed corpus. Claude Sonnet 5.5, validated before acceptance, labels each cited proposition–passage pair; every method then reads the same sealed inputs. Full collection, checker validation, policies, and statistical tests are in Supplement, Section S8.

We use different populations for different questions. All-item accuracy measures performance over the full primary set. A *decisive conflict* contains both a correct and an incorrect proposed answer, so it measures selection where either outcome is available. A *contested* item has at least two distinct proposed answers, whether or not any is correct; cost and replay analyses

use this broader population because a selection decision is still needed. TruthfulQA is reported separately because it tests misconceptions that the corpus rarely addresses.

We also distinguish two baselines. The reported majority vote uses seeded-random tie breaking, whereas the scored rule's own plurality default D uses its deterministic selector. Comparing the final rule with the reported majority therefore includes any effect of changing the tie policy. Comparing it with D measures changes relative to the default used in the formal harm result.

Rules and analysis timing. The original primary rule uses the scoring selector, with a not-enough-information factor of 0.5; the separately tuned ungated comparator uses 0.1. Verified vote instead selects by plurality among verified answers. Verified default first keeps a verified default and otherwise uses verified vote. The original scored rule improved accuracy but missed its pre-specified harm criterion. Verified default first was developed after analyzing the first pool, then specified before testing a second agent pool on the same questions, with some checker labels reused. Its first-pool results are exploratory; its second-pool results test transfer across agent pools, not generalization to unseen questions. Protocols were hash-committed before their respective collections, without a public registry timestamp. New matched-parameter analyses are reported as post hoc and do not replace the original tests.

4.1 Accuracy: overall gains and matched baselines

Over all 1,000 primary items, decisive or not, verify-then-vote raises accuracy from 77.1% for majority vote to 85.0%. The 1,000 primary items yield 526 decisive conflicts, on which majority vote is right 376 times and wrong 150 times. Verify-then-vote corrects 91 of the 150 wrong majorities and overturns 12 of the 376 right ones, raising accuracy from 71.5% to 86.5%. The difference is 0.150 (Newcombe (1998) 95% CI [0.115, 0.186]), with the exact McNemar (1947) test, $p = 3.5 \times 10^{-16}$ (Holm-adjusted 6.9×10^{-16}). The pre-specified superiority null is rejected in favor of verify-then-vote.

Verified vote, the plurality selector, reaches the same 86.5% with 88 corrections and the fewest overturns of any pre-specified evidence-using method, 9. The scoring selector therefore adds no accuracy on this benchmark. Verified evidence supplies most of the gain, and the gate sets the balance between corrections and harmful overturns. The ungated reducer, which reads the same labels, also beats the vote (99 corrected, 23 overturned, 85.9%, $p = 2.0 \times 10^{-12}$). This comparison changes both the gate and the not-enough-information factor, so it does not isolate the gate's effect.

The fixed plurality default D is correct on 386 of the same 526 items (73.4%), against 376 for the seeded-random-tie majority baseline. The scored rule corrects 79 errors and loses 10 correct answers relative to D, reaching the same 455 correct answers (86.5%). Thus 10 of the 79 net gains against the reported majority baseline come from changing the tie default; the remaining 69 are the net gain from verification and selection relative to that fixed default. These matched-default counts (Table 2) are descriptive; the pre-specified primary comparison remains unchanged. Table 1 gives every method.

The gain holds on every source (Table 3). It is largest where the corpus speaks directly to the claim: on FEVER, 28 corrections against 2 overturns, and on SciFact, 33 against 2. On the 43 SciFact claims whose gold is REFUTED, which the fleet tends to accept, accuracy rises from 39.5% to 83.7% with 19 corrections and no overturn. On the TruthfulQA stratum, where the corpus rarely addresses the misconception, verify-then-vote still corrects 9 majorities against 1 overturn (44.9% to 56.5%, Newcombe 95% CI [0.030, 0.197]). The pre-specified non-inferiority criterion therefore holds: it is non-inferior to the vote, and, descriptively, the interval also excludes zero. Both pre-declared sensitivity analyses agree: without PopQA, 61 corrections against 4 overturns ($p = 3.9 \times 10^{-14}$), and with majority ties broken by earliest arrival instead of at random, 90 against 11.

Table 1: Accuracy on the 526 decisive conflicts of P, and paired comparison with majority vote: b corrects a wrong majority, c overturns a right one. †Specified after this analysis; tested on a second pool using the same questions in Table 5.

Method	Model calls per decision	Accuracy	b	c
Majority vote	none	71.5%	—	—
Fixed plurality default D	labels for tied plurality	73.4%	13	3
Verify-then-vote, scored (pre-specified)	checker labels	**86.5%**	**91**	**12**
Verified vote	checker labels	86.5%	88	9
Verified default first†	checker labels	87.3%	87	4
Ungated reducer	checker labels	85.9%	99	23
Evidence-only weights	checker labels	85.7%	100	25
Confidence-weighted vote	none	73.6%	13	2
Best single model (Mistral)	none	69.6%	58	68
Cite vote (unverified)	none	69.0%	29	42
Order-free reducer, unchecked citations	none	63.7%	47	88
Local judge (Mistral 7B)	1 local	69.6%	54	64
Sonnet 5.5 closed-book	1 hosted	90.9%	128	26
Sonnet 5.5 judge	1 hosted	93.9%	131	13
Sonnet 5.5 reader	1 hosted	94.3%	134	14

Table 2: Three distinct accuracy comparisons for the original scored rule on the first pool. All-item accuracy measures the overall gain; the two conflict rows distinguish the reported majority baseline from the rule's own default.

Population	Baseline	Baseline accuracy	Scored rule	Gain
All 1,000 primary items	Majority	77.1%	85.0%	7.9 pp
526 decisive conflicts	Majority	71.5%	86.5%	15.0 pp
Same 526 conflicts	Fixed default D	73.4%	86.5%	13.1 pp

4.2 Harm and what produces it

The scored rule overturns 3.2% of right majorities (12 of 376), with a one-sided 97.5% upper bound of 5.5%, just above the pre-specified 5% ceiling. The scored verified-default-first variant keeps a verified default instead of re-selecting among verified answers and removes half of these overturns (87 corrections against 6). Verified default first over plurality overturns 4 right majorities against 87 corrections, with an upper bound of 2.7%. The second pool tests that rule prospectively (Section 4.4).

Corollary 1 says that verify-then-vote loses a correct default only where some wrong answer is verified. On P the gate's own default winner D, an order-free plurality, was correct and the gated winner wrong on $c_D = 10$ items. All 10 lie among the $F = 56$ items where some wrong answer carries an accepted supports label, and the same holds on T ($c_D = 0$, $F = 4$). The bound holds on every item, as the theorem requires. Of the 12 overturns counted against majority vote, 11 fall among the F items. The twelfth is a plurality tie: random tie-breaking chose correctly, but the deterministic default did not.

Table 3: Verify-then-vote against majority vote by source, on decisive conflicts.

Source	Decisive	Majority	Scored verify-then-vote	b / c	p
FEVER	199	76.9%	89.9%	28 / 2	8.7×10^{-7}
SciFact	154	68.8%	89.0%	33 / 2	3.7×10^{-8}
of which gold REFUTED	43	39.5%	83.7%	19 / 0	3.8×10^{-6}
PopQA	173	67.6%	80.3%	30 / 8	4.7×10^{-4}
TruthfulQA (stratum T)	69	44.9%	56.5%	9 / 1	0.021

What the verified wrong answers are. Two annotators, both Claude Opus 5.5 instances and each blind to the other's labels, classified every one of the 82 supported (incorrect answer, passage) pairs on the 56 items. Each read the proposition against the passage and the gold and assigned one of four classes. The two annotations agree on 79 pairs (Cohen's $\kappa = 0.95$). A third Claude Opus 5.5 pass adjudicated the three disagreements against the class definitions, and the per-pair table with both labels is released with the artifact. An item takes the first class, in the order listed, that any of its pairs has. These are model-attributed categories, not independently validated diagnoses; agreement measures consistency rather than accuracy. The artifact includes blank review forms for all 82 pairs, with passage-relation assessment separated from gold review. No human review has yet been completed.

- *Checker false support* (24 items): the passage does not support the proposition as written, mostly answers of the wrong type (a state given where a country is asked) or non-answers.
- *Entity mismatch* (8): the passage is accurate about a different entity with the same name, and supports the proposition only under that reading. The corpus fact is correct.
- *False or outdated source* (5): the passage states, about the intended entity, an outdated or alternative account.
- *Gold or grading* (19): a spelling or alias the gold list lacks (9), a coarser or alternative answer the passage supports (3), or a source–gold disagreement, in which the passage contradicts the gold label and either can be the one in error (7). We count the last as suspected gold issues.

The 10 lost defaults divide 5, 1, 1, and 3, and the 3 in the last class are all source–gold disagreements. Checker false supports and entity mismatches are both failures of matching the answer to the passage, addressed by the checker and the proposition renderer. False sources are addressed by the corpus, and gold issues by the benchmark.

What the forecast does not establish. The historical harm forecast is not calibrated to checker-caused harm: it counts checks on gold-incorrect answers rather than adjudicated non-entailing passages, and its validation rate excludes false-refute errors that can verify REFUTED answers. The comparison with model-attributed checker errors cannot establish overprediction. The original calculations and calibration requirements are in Supplement, Section S8.2.

Matched gate ablation (post hoc). To isolate the gate, we re-ran the sealed first-pool inputs with it off and on at each of the two not-enough-information factors, holding every other policy field and input fixed. No new model calls were made. At factor 0.1 the gate reduces losses of correct majorities from 23 to 12 but also reduces corrections from 99 to 89; accuracy changes from 85.9% to 86.1%. At factor 0.5 it reduces losses from 33 to 12 and corrections from 96 to 91, raising accuracy from 83.5% to 86.5% (Table 4). Thus the lower harm is present in both matched comparisons, with a cost in corrections. These descriptive analyses supplement the original tests; they were not pre-specified. The default is fixed within each comparison, but its tied-plurality choices can differ between factors.

Table 4: Matched gate ablation on the same 526 decisive primary conflicts. Corrections and losses are relative to the reported majority baseline. Within each factor, only the gate changes.

NEI factor	Gate	Accuracy	Corrections	Losses
0.1	off	85.9%	99	23
0.1	on	86.1%	89	12
0.5	off	83.5%	96	33
0.5	on	86.5%	91	12

4.3 What the passage controls establish

The controls in Table 1 separate what produces the gain. Citations alone do not: the order-free reducer weighting the same unverified citations overturns more right majorities than it corrects (47 against 88), and a vote among citing agents scores 69.0%. Checking the citations changes this. Every pre-specified evidence-using configuration in Table 1 rises above 85%. The two pre-specified gated rules reduce harmful overturns relative to the separately tuned ungated methods. Scored verify-then-vote and verified vote tie on accuracy (9 against 9 between them, $p = 1.0$); verified vote overturns the fewest right majorities among the pre-specified evidence-using methods. Evidence-only weights match the full ungated reducer (6 against 5), so the checker's labels carry the signal and the remaining score features refine ties. A companion pre-specified benchmark isolates the same point from the other side. On 600 questions from the same four datasets, the same five agents answered from memory with no corpus, so there was nothing to verify. On its 293 decisive conflicts the order-free reducer scored 58.4% against majority vote's 62.8% (45 against 58, exact McNemar $p = 0.24$, Newcombe 95% CI $[-0.111, 0.023]$), and its pre-specified 10-point non-inferiority margin was not established. On OVERTURN, where there are passages to verify, verify-then-vote gains 15 points over the vote.

The gain depends on passages under ablation. A control run after the confirmatory analysis tests whether the checker's supporting labels persist without the cited text or with unrelated text. We took the 118 witness pairs that licensed overturns on P and 118 seeded-random supported pairs that did not. The same checker labeled each answer again, once with the passage withheld and once with the passage replaced by an unrelated passage from the same source. In each condition it returned supports on none of the 236 pairs (two-sided 95% Wilson upper bound 1.6%). Not one of the 118 licensing pairs among them was supported (upper bound 3.2%). The checker labeled every one not-enough-information. Re-deciding P with these labels in place of the witnesses' labels leaves 15 corrections and 2 overturns, 74.0% against the majority's 71.5%. The substitution removes 76 of the 91 corrections and 10 of the 12 overturns. Of the remaining corrections, 13 involve no overturn of the fixed default and arise from how it breaks plurality ties. These controls establish dependence on the cited passages in the tested path, not the absence of parametric-knowledge influence when relevant passages are present.

Against the references, verify-then-vote closes 66.9% of the gap between majority vote and the Sonnet 5.5 judge (bootstrap 95% CI [55.8%, 77.9%]). The judge, the reader, and the closed-book model use outside knowledge and a much larger model at every decision. Verify-then-vote

Table 5: Second agent pool, pre-specified: decisive conflicts on P (547; majority right on 400) and on the TruthfulQA stratum T (68). Bound: one-sided 97.5% Clopper–Pearson upper bound on the rate of overturned right majorities.

Method	Accuracy	b	c	p	Bound
Majority vote	73.1%	—	—	—	—
Verified default first (primary)	**85.9%**	**73**	**3**	1.9×10^{-18}	**2.2%**
Verify-then-vote, scored	86.3%	80	8	4.6×10^{-16}	3.9%
Scored, verified default first	86.3%	77	5	1.2×10^{-17}	2.9%
Verified vote	85.9%	76	6	1.6×10^{-16}	—
Ungated reducer	85.7%	92	23	5.8×10^{-11}	8.5%
Best single model (EXAONE 3.5)	79.3%	84	50	0.004	—
Sonnet 5.5 judge	93.6%	122	10	1.2×10^{-25}	—
Sonnet 5.5 reader	94.3%	128	12	1.2×10^{-25}	—
TruthfulQA stratum T (majority 48.5%)					
Verified default first	50.0%	2	1	1.0	—
Verify-then-vote, scored	51.5%	7	5	0.77	—
Ungated reducer	35.3%	10	19	0.14	—

uses that model only to label passages and decides with a frozen rule. A judge of the agents' own size mostly follows the wrong majority. Given the same answers, confidences, and passages, the local Mistral judge follows it on 87 of 150 items and scores 69.6%. Verify-then-vote beats it by 110 decisions to 21 ($p = 9.4 \times 10^{-16}$). A judge weighs five answers at once; the gate asks its checker only to label one passage against one proposition, a task validated before use. The judge's errors concentrate where the gate's do. It is wrong on 17 of the 56 questions with a verified wrong answer, including 9 of the 19 gold issues, and on 15 of the other 470 (odds ratio 13.2, $p = 4.5 \times 10^{-10}$). Paired with the scored rule on the 526 decisive conflicts, the judge is right where the rule is wrong on 50 and wrong where the rule is right on 11. Fourteen of its 50 wins fall on questions with a verified wrong answer, including 5 of the rule's 10 lost defaults, so the judge does rescue some of the rule's losses. The other 36 are questions on which no answer was verified and the rule kept a wrong default. On the second pool, verified default first closes 63% of the gap to the judge.

4.4 Second agent pool on the same questions

The second pool tests verified default first on newly collected responses from five other model families to the same 1,100 questions, reusing an existing checker label whenever the proposition–passage pair is unchanged. The rule and its analysis were specified after the first-pool analysis but before this collection. Its primary tests require both superiority to majority vote and a one-sided 97.5% upper bound below 5% on overturns of correct majorities. This is a prospective test across agent pools, not on unseen questions; the full protocol and repeated original-rule tests are in Supplement, Section S8.

The new pool reaches 547 decisive conflicts on P, on which majority vote is right 400 times. Verified default first corrects 73 of the 147 wrong majorities and overturns 3 of the 400 right ones, from 73.1% to 85.9% ($p = 1.9 \times 10^{-18}$; Newcombe 95% CI [0.099, 0.158]). Its overturn rate is 0.75% with a one-sided 97.5% upper bound of 2.2%, so both steps reject, and harm below 5% is established for verified default first on this pool. Removing the 15 items whose harm cases informed the rule leaves 73 against 3 (bound 2.2%), and grading with a frozen alias list leaves the same counts. Every lost default lies among the 46 items with a verified wrong answer, as Corollary 1 requires. The gain holds on every source: 22 against 2 on FEVER, 21 against 0 on SciFact, and 30 against 1 on PopQA.

Three of the four original OVERTURN criteria are met on the second pool: superiority of the

scored and ungated configurations to majority vote, and the scored rule's harm-rate criterion. The accuracy gains hold on the new agent pool, and the harm-rate criterion, which the first pool narrowly missed, is satisfied on this pool. The fourth, non-inferiority on TruthfulQA, is inconclusive. The scored rule beats majority vote with 80 corrections against 8 ($p = 4.6 \times 10^{-16}$, Holm-adjusted 9.2×10^{-16}) and the ungated reducer does so with 92 against 23. The scored rule's overturn rate has an upper bound of 3.9%, below the ceiling; across both pools it overturns 20 of 776 right majorities. Treating these as a single binomial sample gives a 3.95% upper bound. The secondary Bonferroni sensitivity analysis permits different pool-specific rates and dependence between pools, giving 5.01%, just above the ceiling; it retains within-pool binomial assumptions (Supplement, Section S8.3). The scored gated configuration has fewer overturns than the separately tuned ungated configuration (8 versus 23); because their evidence factors differ, this is a comparison of configurations, not an isolated gate effect. On the TruthfulQA stratum, where the corpus rarely speaks to the misconception, the gated rules stay level with the vote (2 against 1 and 7 against 5), and non-inferiority is not shown at 68 items. The ungated reducer, by contrast, overturns more right majorities than it corrects (10 against 19). Effect sizes on P agree across pools within their confidence intervals.

4.5 Audit, replay, and cost

Every OVERTURN decision is a replay of a sealed input. Each of the 1,100 items was reduced twice under each evidence policy and the outputs were byte-identical in every case. An independent re-implementation of the reducer's selection matched the binary on all 7,910 slots. Because the checker's labels are sealed, we can replay a decision after removing an agent. For each of the 702 contested items, we omitted one agent and selected from the remaining four, then restored it and omitted the next. These five variants per item gave $702 \times 5 = 3{,}510$ decisions. Replaying them took 23 s with no new model call, and 9.4% selected a different answer from the original decision. The same analysis for the Sonnet judge needs five new calls, about \$0.071 and 19 s per item. Re-asking the judge on 200 items reproduced 97.5% of its own decisions.

Each overturn is also small to audit. The gate's trace names the verified answer and its witnesses. An overturn on P is licensed by a median of one labeled passage of about 950 characters (maximum four), against a median of 3,649 tokens of context behind each judge decision. The reducer takes 6.0 ms per decision (95th percentile 12.6 ms).

The model cost is the checker's, and it is counted only where a label can change a decision. Agents cited 4,742 distinct (proposition, passage) pairs across all 1,100 items, each labeled once. Of these, 2,739 fall on the 622 contested items of P (4.4 per decision) and 440 on the 80 contested items of T (5.5 per decision); the other 1,563 fall on unanimous items, where no label can change the answer. A further 2,506 labels on gold-answer propositions measured how often the retrieved passages supported the gold (Supplement, Section S6). They are a diagnostic and enter no decision or cost. At a measured \$0.0048 per label, verify-then-vote costs \$0.021 per contested decision on P, 1.5 times one judge call (\$0.014), so the judge is cheaper for a first decision. The stored labels make every later decision cheaper. All evidence-using methods and every counterfactual above re-decide from the same labels at no further model cost. This reuse extends to policies restricted to the existing sealed inputs and accepted labels; new answers, evidence, proposition renderings, or checker identities can require additional checking, and each label remains individually auditable. The labels also carry over to new agents. A label is keyed by its proposition and passage, so when the second pool cited a pair the first pool had cited, its sealed label was reused. In all, 49% of the second pool's cited checks were already labeled. Its 634 contested decisions on P read 4.4 labels each, as before. Priced at each label's original call, the cost is again \$0.021 per decision; the new calls cost \$0.011, 0.83 of one judge call (\$0.014), so a second pool's first decisions already cost less than a judge's. Verified vote, verified default first, the scored rule, and the ungated reducer read the same labels and share this cost.

4.6 Replay and tamper evidence

The replay tests cover five synthetic topics with 100 agents each (seed 42). All 500 in-memory recompilations and 100 input-order permutations preserved their respective certificates and decisions. Four builds—native macOS arm64, translated macOS x86-64, containerized Linux arm64, and translated containerized Linux x86-64—produced 20 dumps that were byte-identical per topic, and all 80 cross-replays passed. These builds span two instruction sets, two operating systems, and two compiler patch releases on one physical host; they do not establish agreement for every possible input or platform.

Nine constructed forgeries, replayed on all four builds, were rejected identically in all 36 trials. For example, editing a claim and recomputing its root hash still fails if its retained identifier no longer matches its content. Detection is relative to trusted outputs or a certificate held independently of the bundle: an attacker who replaces all copies of that commitment can recompute the hashes. The full replay matrix, mutation inventory, and further field-level tests are in Supplement, Section S10; Section 5.2 states the scope of these results.

5 Discussion

5.1 Related work

Aggregation. Self-consistency votes over sampled reasoning paths (Wang et al., 2023), and debate has agents revise answers after reading each other (Du et al., 2024). Kaesberg et al. (2025) find voting stronger than consensus on reasoning tasks. Recent studies find that majority voting accounts for most of debate's gains (Choi et al., 2025; Zhang et al., 2025) and improve on it with confidence- and diversity-aware debate (Zhu et al., 2026) or second-order aggregation across models (Ai et al., 2026). Cemri et al. (2025) catalog 14 failure modes of multi-agent systems. These works improve the accuracy of the combined decision. We focus on recording the decision so that it can be audited. OVERTURN agrees that majority vote is a strong baseline once answers are constrained, and shows a deterministic way past it: overturning only through verified answers corrects 61% of wrong majorities. Zheng et al. (2026) study multi-agent reliability through Byzantine fault tolerance; we target auditability of the decision rather than consensus under Byzantine faults.

LLMs as judges. Model judges track human preferences but show position, verbosity, and self-enhancement biases (Zheng et al., 2023). Later work quantifies a dozen bias types (Ye et al., 2025), shows position bias varies with judge, task, and candidate quality gap (Shi et al., 2025), and surveys reliability concerns (Gu et al., 2024). Our design confines model judgment to the narrowest task, labeling one passage against one proposition, where it can be validated in advance. The label is sealed, the checker's identity is hashed into the policy, and the gate ties every loss of a correct default to a verified wrong answer that can be inspected. On OVERTURN this recovers two thirds of a Sonnet judge's gain over the vote while every decision stays a replayable function.

Determinism. Nominally deterministic LLM settings still vary across runs (Atıl et al., 2025), and results shift with batch size, GPU count, and precision because floating-point addition is non-associative (Yuan et al., 2025). Batch-invariant kernels remove much of this at the serving layer (He and Thinking Machines Lab, 2025). Our approach complements this with the purely functional deployment model of Nix (Dolstra et al., 2004): record non-deterministic outputs as content-addressed inputs and make the downstream function pure. Exact logging with any deterministic reducer would also replay (Supplement, Section S9); sealing adds typed inputs, tamper evidence, and conditions for justified selection.

Claim extraction and verification. Ullrich et al. (2025) benchmark claim extraction for fact-checking, and Kolli et al. (2025) verify claims with knowledge graphs, LLMs, and search. Either kind of system can serve as an artifact producer at the arbitration boundary, for lowering or for

support checking, and their benchmarks are the natural yardstick for our rule-based lowering (Supplement, Section S3). Verify-then-vote also applies retrieval-grounded verification of the kind FEVER and SciFact pioneered (Thorne et al., 2018; Wadden et al., 2020) to aggregation: the verifier licenses an overturn rather than producing the answer.

Knowledge conflicts. Work on conflicts between a model's parametric knowledge and retrieved context (Xu et al., 2024) finds that models adopt incorrect retrieved content readily, more so the less confident their prior (Wu et al., 2024). Verify-then-vote takes the conflict out of the model: agents may disagree with each other and with the corpus, and only a checked label on a cited passage can move the decision. False sources are the residue of this conflict that a correct checker cannot remove, and Proposition 1 counts them.

Verifiers, attribution, and crowd aggregation. Verifier-guided selection reranks sampled solutions with a trained outcome or process verifier (Cobbe et al., 2021; Lightman et al., 2024). Verify-then-vote keeps the vote as the default and lets a verifier act only through a cited passage, so each overturn can be checked against sealed text. Attribution work defines and measures whether generated text is supported by its cited source (Rashkin et al., 2023; Gao et al., 2023), and compact grounding checkers (Tang et al., 2024) are candidates for a local checker once they label refutation as well as support under our unchanged claim representation (Supplement, Section S8). Classical label aggregation weights annotators by estimated error rates (Dawid and Skene, 1979), and the surprisingly-popular rule overturns a majority using respondents' predictions of others' answers (Prelec et al., 2017). In verify-then-vote, the overturn signal is external evidence, checked once and sealed.

5.2 Scope

The results hold under the following conditions. Accuracy gains require agents that cite passages the checker can read. On OVERTURN these come from a fixed corpus. Open web citations resolve and attach to their claims far less often (Supplement, Section S6), which limits what the checker can verify. The harm bound is exact for the rule's deterministic default and states harm in units of verified wrong answers. Its practical value is therefore set jointly by the checker's error rate, entity resolution, the correctness of the corpus, and, in evaluation, the completeness of the gold. The checker, an LLM, is part of the trusted computing base. The classification of OVERTURN's verified wrong answers rests on two same-family model annotators and model adjudication, without independent human validation; their agreement does not establish attribution accuracy. Verified default first was specified after the first pool's analysis and is tested on one further pool using the same questions and partly reused checker labels, so this is not independent confirmation on unseen questions. The pooled sensitivity analysis allows dependence between pools but still relies on a binomial model within each pool; it does not establish harm below the pre-specified ceiling (Supplement, Section S8.3). TruthfulQA non-inferiority was established on the first pool and remains inconclusive on the second. The historical harm forecast uses gold-incorrect-answer exposure rather than adjudicated non-entailing passages and omits the false-refute validation channel. It is not a calibrated estimate of checker-caused harm; model-based attribution cannot repair those definition gaps. The passage ablation demonstrates dependence on supplied text, not the absence of parametric-knowledge influence. Tamper evidence is relative to a certificate the verifier holds independently of the bundle and assumes no relevant hash collision. The reference certificate is only 64 bits (Supplement, Section S2); the tests do not establish cryptographic security against an adversary searching for collisions. The benchmark uses two pools of five open-weight models, one checker, one seed, and one machine. The cross-build replay matrix ran on one physical host with both x86-64 builds binary-translated, demonstrating replay on the tested instruction sets, operating systems, and C libraries. Captures attest the bytes received rather than origin integrity. Conflict detection follows declared predicate arity and exclusivity, and quantifier scope, temporal qualification, and complex negation lie outside the lookup. The protocol was frozen and hash-committed before data collection,

but no public registry timestamp exists, so its timing rests on the study records. A *verified* label also means only that the checker accepted the cited passage as supporting the answer; it can mislead if the checker or the passage is wrong, so readers should not treat it as proof the answer is true.

5.3 Future work

Natural next steps are human-validated reading of the verified wrong answers and of the gold, and a local three-way checker validated to the hosted checker's thresholds, which would make every decision free of hosted calls. Others are verify-then-vote over open web evidence and replay of live-ingestion histories with streaming retraction.

6 Conclusion

Verify, then vote improves majority accuracy when agents cite passages that a checker can assess. Verification supplies the main signal: the simpler plurality selector matches the scored rule's accuracy, and the passage controls remove most corrections. Keeping a verified default reduces harmful overturns, but the original scored rule misses its pre-specified harm criterion. The second-pool result supports transfer across agents on shared questions; unseen-question generalization and independent human validation of the failure attribution remain open.

The formal guarantee makes failures inspectable, not necessarily rare. The practical contribution is an explicit selection policy whose passage evidence can be audited and reused for later decisions over existing sealed inputs. A strong judge remains more accurate and cheaper for a first decision; verify-then-vote is useful when this policy control and evidence reuse matter. The artifact and Supplement, Section S7 make the measured tradeoffs available for re-analysis.

Artifact Availability

Code, Lean proofs, sealed inputs, study protocols, and analysis scripts accompany the paper at `https://github.com/grainulation/papers/tree/main/papers/verify-then-vote`. The artifact includes the matched gate ablation, validation-channel audit, blank human-review packet, and portable source-reconstruction checks. Supplement, Section S7 and artifact README map results to their reproduction commands. Protocol freezes record when methods and analyses were specified; they are not prior publication versions. Exact public-redaction mappings preserve the relationship to the frozen study records.

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